№1

LaTeX

2024

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LaTeX TeX Live.

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- TeX Live
- LaTeX
- LaTeX-
- LaTeX-

TeX Live

TeX Live .

Ubuntu `sudo apt install texlive-full`.

`pdflatex --version`.

```
Installing the following packages:
texlive
By installing, you accept licenses for the packages.
texlive v2025.20250902.0 already installed.
 Use --force to reinstall, specify a version to install, or try upgrade.

Chocolatey installed 0/1 packages.
 See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).

Warnings:
 - texlive - texlive v2025.20250902.0 already installed.
 Use --force to reinstall, specify a version to install, or try upgrade.
PS C:\Users\sidra\Desktop> pdflatex --version
pdfTeX 3.141592653-2.6-1.40.28 (TeX Live 2025)
kpathsea version 6.4.1
Copyright 2025 Han The Thanh (pdfTeX) et al.
There is NO warranty.  Redistribution of this software is
covered by the terms of both the pdfTeX copyright and
the Lesser GNU General Public License.
For more information about these matters, see the file
named COPYING and the pdfTeX source.
Primary author of pdfTeX: Han The Thanh (pdfTeX) et al.
Compiled with libpng 1.6.46; using libpng 1.6.46
```

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```
PS C:\Users\sidra\Desktop> notepad first.tex
PS C:\Users\sidra\Desktop> pdflatex first.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./first.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file first.aux.

[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}] (./first.aux)
 )
kpathsea: Running mktexpk --mfmode / --bdpi 600 --mag 1+0/600 --dpi 600 ecrm1000

The command name is C:\texlive\2025\bin\windows\mktexpk
mf-nowin.exe --progname=mf --base=mf \mode:=ljfour; \mag:=1+0/600; nonstopmode; input ecrm1000;
This is METAFONT, Version 2.71828182 (TeX Live 2025) (preloaded base=mf)

(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/ecrm1000.mf
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exbase.mf)
```

. **2:** pdflatex –version

LaTeX-

.tex

.tex LaTeX- , :

- \documentclass{article};
- babel amsmath;
- \title, \author, \date \maketitle.

Hey world!
This is a first document.

LaTeX- `pdflatex laboratorio1.tex`.

`laboratorio1.pdf`, .

```
PS C:\Users\sidra\Desktop>
PS C:\Users\sidra\Desktop> notepad exercise1.tex
PS C:\Users\sidra\Desktop> pdflatex exercise1.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercise1.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file exercise1.aux.

[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}]
(./exercise1.aux) )
kpathsea: Running mktexpk --mfmode / --bdpi 600 --mag 1+0/600 --dpi 600 ecti1000
```

. **4:** laboratorio1.tex

PDF- .

Hey world!

This is a first document with multiple paragraphs.

This is the second paragraph with different spacing. Look at how spaces are handled.

This is a paragraph with hard spaces that won't break.

Adding special characters: $100, 25%, C# programming, and LaTeX_doc.

Testing **bold text** and *italic text*.

LaTeX:

- inline- `$...$`;
- display- `\[...\]`.

```
PS C:\Users\sidra\Desktop> notepad structure.tex
PS C:\Users\sidra\Desktop> pdflatex structure.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./structure.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size12.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file structure.aux.

[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}]
(./structure.aux) ) <c:/Users/sidra/.texlive2025/texmf-var/fonts/pk/ljfour/jkna
ppen/ec/ecrm1000.600pk>
kpathsea: Running mktexpk --mfmode / --bdpi 600 --mag 1+0/600 --dpi 600 ecrm0700

The command name is C:\texlive\2025\bin\windows\mktexpk
mf-nowin.exe --progname=mf --base=mf \mode:=ljfour; \mag:=1+0/600; nonstopmode; input ecrm0700;
```

. 6:

$E = mc^2$ LaTeX.

A sentence with inline mathematics: $y = mx + c$.
A second sentence with inline mathematics: $5^2 = 3^2 + 4^2$.
A second paragraph containing display math:

$$y = mx + c$$

See how the paragraph continues after the display.
Simple formulas: a^b and a_b.
Greek letters: $\alpha, \beta, \gamma, \Gamma$.

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- TeX Live;
- LaTeX- ;
- (`\documentclass`, `\section`, `\maketitle`);
- ;
- PDF.

LaTeX .